

# **Project Name :- Insurance Risk Assessment & Claims Analytics Platform**

## **1. Project Overview**

The insurance industry generates large volumes of customer and policy data every day. Insurance providers rely on this data to evaluate customer risk, determine premium pricing, and minimize financial losses resulting from insurance claims. However, identifying high-risk customers and understanding the factors contributing to claim occurrences can be challenging without a structured analytical approach.

This project develops an end-to-end Insurance Risk Assessment & Claims Analytics Platform using Python, SQL, and Power BI. The objective is to transform raw insurance policy data into meaningful business insights that help insurers make informed underwriting, pricing, and risk management decisions. The project analyzes customer demographics, vehicle characteristics, safety features, and regional information to identify patterns associated with insurance claims.

## **2. Business Problem Statement**

Insurance companies often face increasing claim costs while attempting to maintain competitive premium pricing and customer satisfaction. Traditional methods of evaluating policy risk may overlook hidden relationships between customer characteristics, vehicle features, and claim behavior.

Some of the major business challenges include:

- Difficulty identifying high-risk customers before policy approval.
- Limited understanding of how vehicle characteristics influence claim occurrence.
- Inconsistent premium pricing across customer segments.
- Lack of visibility into the impact of safety features on insurance claims.
- Regional variations in claim frequency that are difficult to monitor.
- Limited executive reporting for strategic decision-making.

Without a comprehensive analytics solution, these challenges can lead to higher claim costs, inefficient underwriting, and reduced profitability.

### **3. Business Objectives**

The primary objectives of this project are to:

- Analyze historical insurance policy data to identify claim patterns.
- Understand customer characteristics associated with higher insurance claim risk.
- Evaluate how vehicle attributes influence claim occurrence.
- Measure the effectiveness of safety features in reducing insurance claims.
- Analyze regional trends and geographical claim behavior.
- Develop interactive dashboards for business monitoring.
- Generate data-driven recommendations to improve underwriting and premium pricing.

### **4. Industry Background**

Insurance companies assess risk before issuing policies to determine appropriate premium amounts. Accurate risk assessment helps insurers reduce financial losses while ensuring fair pricing for customers.

Modern insurance organizations increasingly rely on data analytics to:

- Identify customers with higher probabilities of filing claims.
- Detect vehicle categories associated with increased insurance risk.
- Evaluate the effectiveness of vehicle safety technologies.
- Improve pricing strategies based on customer risk profiles.
- Monitor regional claim trends for better resource allocation.

An end-to-end analytics platform enables decision-makers to move from reactive reporting to proactive risk management. By leveraging historical insurance data, organizations can uncover hidden patterns in customer behavior, vehicle characteristics, and claim trends that are not easily identifiable through traditional reporting methods. These insights enable insurance

providers to make evidence-based decisions, enhance operational efficiency, and develop more personalized risk assessment strategies.

## **5. Dataset Overview**

The dataset consists of 58,592 insurance policy records, where each row represents an individual insurance policy.

The dataset includes information related to:

### **Customer Information**

- Customer Age
- Customer Age Group
- Subscription Length
- Subscription Tenure Group

### **Vehicle Information**

- Vehicle Age
- Vehicle Age Group
- Vehicle Segment
- Fuel Type
- Transmission Type
- Engine Type

### **Safety Features**

- NCAP Rating
- Safety Feature Score
- Airbags
- Electronic Stability Control (ESC)
- TPMS

- Parking Sensors
- Parking Camera

#### Regional Information

- Region Code
- Region Density Group

#### Insurance Information

- Policy ID
- Claim Status
- Claim Status Label

## 6. Project Scope

This project focuses on analyzing insurance claim behavior from multiple business perspectives.

The scope includes:

- Data Quality Assessment & Validation
- Exploratory Data Analysis (EDA)
- Advanced SQL Analytic
- Customer Risk Analysis
- Vehicle Risk Analysis
- Insurance Claim Analysis
- Vehicle Safety Analysis
- Regional & Policy Analysis
- Executive Business Insights
- Performance Monitoring using KPIs
- Interactive Dashboard Development

- Business Recommendation Generation
- Data-Driven Decision Support

## 7. Key Business Questions

The project addresses several important business questions, including:

### Customer Analysis

- Which customer age groups exhibit the highest insurance claim rates?
- Which customer profiles require premium adjustments?
- Which customer segments should receive targeted retention strategies?

### Vehicle Analysis

- Which vehicle segments generate the highest claim rates?
- Does vehicle age influence insurance claim frequency?
- Which fuel and engine types present greater insurance risk?

### Safety Analysis

- Do higher NCAP ratings reduce insurance claim frequency?
- Does Electronic Stability Control (ESC) contribute to lower claims?
- How do airbags and safety feature scores affect claim occurrence?

### Regional Analysis

- Which geographical regions experience higher claim rates?
- Does population density influence insurance claims?
- How does policy subscription tenure affect claim behavior?

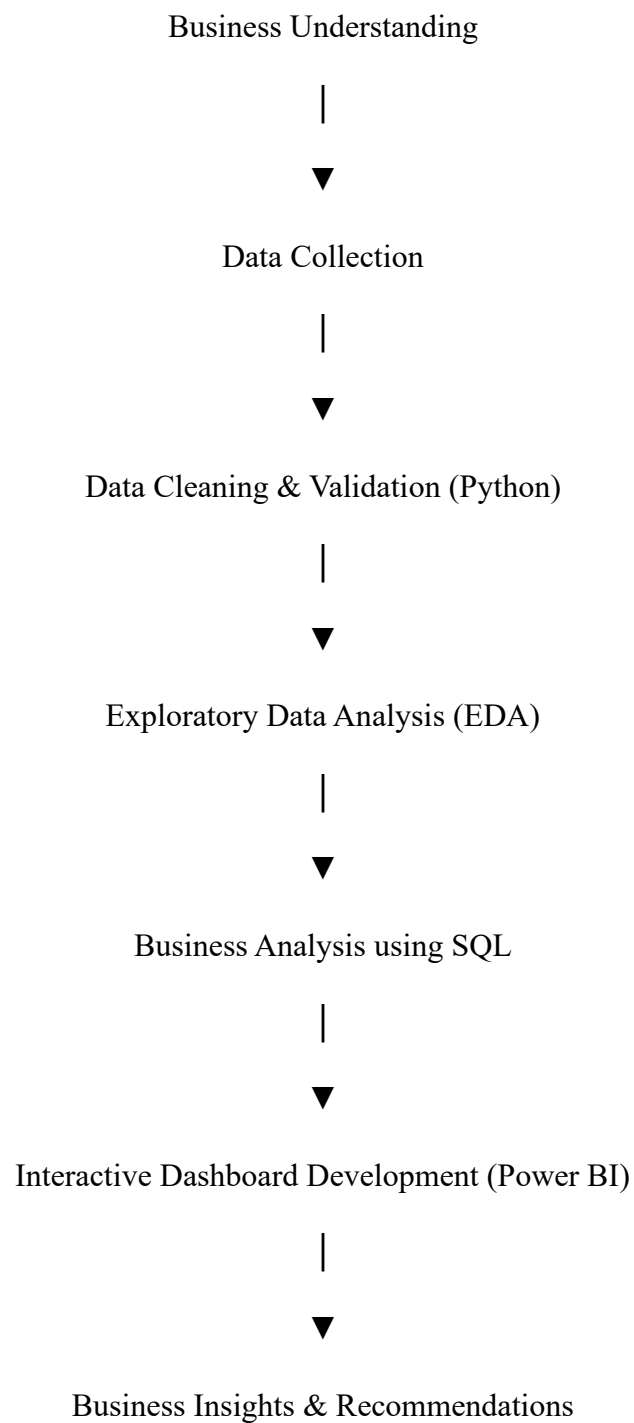
### Executive Insights

- Which customer, vehicle, and regional combinations represent the highest overall insurance risk?

- Which customer segments should receive stricter underwriting or premium adjustments?

## 8. Project Workflow

The project follows a complete end-to-end data analytics workflow:



## 9. Expected Business Outcomes

The insights generated from this project are expected to help insurance companies:

- Improve underwriting accuracy by identifying high-risk policyholders.
- Optimize premium pricing based on customer and vehicle risk profiles.
- Reduce claim-related financial losses.
- Promote safer vehicles through data-driven pricing strategies.
- Identify regional areas requiring enhanced risk monitoring.
- Support executive decision-making through interactive dashboards and analytical reports.

## 10. Key Performance Indicators (KPIs)

The project focuses on monitoring the following KPIs:

| KPI                          | Description                                 |
|------------------------------|---------------------------------------------|
| Total Policies               | Total number of insurance policies analyzed |
| Total Claims                 | Total number of claims filed                |
| Claim Percentage             | Percentage of policies resulting in claims  |
| Average Customer Age         | Mean age of insured customers               |
| Average Vehicle Age          | Mean age of insured vehicles                |
| Average Safety Feature Score | Overall vehicle safety level                |
| Average NCAP Rating          | Average crash safety rating                 |

|                            |                                      |
|----------------------------|--------------------------------------|
| Regional Claim Rate        | Claim percentage by region           |
| Vehicle Segment Claim Rate | Claim percentage by vehicle category |

11. Tools & Technologies

| Technology | Purpose                                                  |
|------------|----------------------------------------------------------|
| Python     | Data cleaning, validation, and exploratory data analysis |
| Pandas     | Data manipulation and preprocessing                      |
| Matplotlib | Exploratory data visualization                           |
| MySQL      | Business analysis using SQL queries                      |
| Power BI   | Interactive dashboard development                        |
| GitHub     | Project documentation and version control                |

12. Deliverables

The project produces the following deliverables:

- Cleaned and validated insurance dataset.
- Python notebook for data validation and exploratory analysis.
- SQL database and business analysis scripts.
- Advanced SQL analytics using CTEs and window functions.
- Interactive Power BI dashboard.
- Business Understanding Report.
- GitHub documentation with project assets and source code.